Madhiyougi Jain

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Education

UMass Amherst, Manning College of Information & Computer Sciences — Amherst, MA

B.S. Informatics (Data Science concentration) and Psychology — *Expected May 2027*

- Coursework: Data Structures, Machine Learning, Databases

Projects

NYC Collisions: Injury Risk Watchlist | *Python, scikit-learn, DuckDB* | Live | GitHub — 2025 – 2026

- Gradient-boosted classifier scores a crash's injury risk from what was involved, not which site; on **121,970** held-out 2025–26 crashes, ROC-AUC **0.79** vs. **0.54** for a borough-hour base rate, Brier skill **+26.9%**.
- Ranked **1,093** intersections by the lower 95% bound of injury rate above their crash mix; in a backtest the ranking persists (Spearman **0.26**, $p = 0.0005$) but only ties a no-model sort.
- Scored **413,780** crashes with five rolling-forward fits so no crash is scored by a model that trained on it: **102** sites clear $z = 1.96$ where chance predicts about 27.
- Weekly GitHub Actions job re-pulls a 30-day overlap from NYC Open Data and dedupes on `collision_id`, rebuilding **637,256** crashes as Parquet behind a DuckDB/Streamlit dashboard; **318** tests.

Hall Check: Dining-Hall Occupancy | *Python, PyTorch (YOLO), Docker, Linux* | GitHub — In progress

- Person counter for five public UMass dining-hall cameras: one frame every 2 min held in memory, YOLO11n counts people whose feet fall inside an ROI polygon, and the frame is never stored; CI fails the build on pixel-writing calls or any detector save flag not pinned off.
- Daily drift alarm compares each camera's last 24 h with its trailing 14 days slot by slot, firing only when busy slots shift by 40% and 4+ people together; an outage reports as missing data, not drift.
- Fixed the evaluation before collecting data: blind hand counts with MAE and signed bias reported apart, an accuracy report that fails until 150+ labels include 10+ per hall, meal, and lighting, and a forecast check that fails unless the 30-min model beats persistence and same-slot baselines.
- Swapped PyTorch's default CUDA build for the CPU wheel, cutting the worker image by over 30%; CI fails the build if CUDA libraries return. **228** worker tests.

When Average Effects Lie: Uplift Modeling | *Python, scikit-learn, statsmodels* | Write-up | GitHub — 2026

- Trained T-, S-, and DR-learners on the **64,000**-customer Hillstrom randomized trial, picking by 5-fold cross-fitted Qini on training rows; picking on the reporting split would have quoted a men's-email Qini of 19.3, not **2.6**.
- Found what the women's email's **+4.5pp** average visit lift hid: **+7.3pp** for prior women's-merchandise buyers vs. **+1.1pp** for the rest (Qini **60.4**), with Benjamini–Hochberg control across the interaction tests.
- Set the send threshold at cost break-even (assumed \$2/visit, \$0.06/email): per 1,000 customers, targeting adds **\$16.95** (95% CI [\$2.99, \$30.60]) and sends **415** fewer emails. For the men's email the same analysis said send to everyone, because its targeting gain could not be told from zero.

Experience

Somneuro Lab, UMass Amherst — Amherst, MA

Undergraduate Research Assistant — *Sep 2026 – Present*

- Onboarded (CITI certified) to the lab's wrist-actigraphy (GGIR) and polysomnography data from studies of sleep and memory consolidation.

Spiritual Data — Remote

AI & Psychology Research Intern — *May 2025 – Aug 2025*

- Built an automated harness that benchmarked a proprietary model against baselines over repeated runs.
- Designed evaluation splits and error slices to find failure modes; wrote the report that informed model selection.

Technical Skills

Languages: Python, SQL, Java, TypeScript, R
ML: scikit-learn, Ultralytics YOLO; gradient boosting, calibration, uplift modeling
Statistics: statsmodels; permutation tests, FDR control, bootstrap CIs
Data: pandas, DuckDB, PostgreSQL
Systems & Tools: Linux, Docker, GitHub Actions (CI/CD), Git, Claude Code (AI-assisted development)